

Astragalus membranaceus polysaccharides potentiate the growth-inhibitory activity of immune checkpoint inhibitors against pulmonary metastatic melanoma in mice

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ABSTRACT

Astragalus membranaceus (*A. membranaceus*) is commonly used in various herbal formulations to treat several human and animal diseases. Polysaccharides, which are the major bioactive components in the *A. membranaceus*, exhibit various bioactive properties. However, the ability of *A. membranaceus* polysaccharides (APS) to activate the mucosal immune response has not been examined. We examined the effect of intranasal administration of APS on mucosal immune cell activation and the growth-inhibitory activity against pulmonary metastatic melanoma in mice by combination treatment with immune checkpoint blockade. The intranasal treatment of APS increased the number of lineage⁺CD11c⁺ dendritic cell (DCs) in the mesenteric lymph nodes (mLN) through the upregulation of CC-chemokine receptor 7 expression. Moreover, intranasal treatment of APS activated DCs, which further stimulated natural killer (NK) and T cells in the mLN. The APS/anti-PD-L1 antibody combination inhibited the pulmonary infiltration of B16 melanoma cells. The depletion of NK cells and CD8 T cells in mice mitigated the anti-cancer effect of this combination, thereby highlighting the critical role of NK cells and CD8 T cells in mediating anti-cancer immunity. These findings demonstrated that APS could be used as a topical mucosal adjuvant to enhance the immune check point inhibitor anti-cancer effect.

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1. Introduction

Astragalus membranaceus (*A. membranaceus*), which is the most widely used herbal medicine in the traditional medicines of China and Korea [1–5], has been used as an immunomodulator for the treatment of fatigue, diarrhea, anorexia, and viral infections [6]. Previous study

Abbreviations: *A. membranaceus*, *Astragalus membranaceus*; APS, *Astragalus membranaceus* polysaccharide; LN, lymph node; PD-L1, programmed cell death-ligand 1; mAbs, monoclonal antibodies; PD-1, programmed cell death-1; DC, dendritic cell; NK, natural killer; APCs, antigen-presenting cells; CTL, cytotoxic T lymphocytes; mLN, mediastinal LN; GPC, gel permeation chromatography; PDI, polydispersity index; GC/MS, gas chromatograph/mass spectrometer; FT-IR, fourier transform infrared; LPS, lipopolysaccharide; IFN, interferon; MHC, major histocompatibility complex; TNF, tumor necrosis factor; DAPI, 4',6-diamidino-2-phenylindole; ELISA, Enzyme-linked immunosorbent assay; BAL, bronchoalveolar lavage; IL, interleukin; H&E, Hematoxylin and eosin; CCR7, CC-chemokine receptor 7; TLRs, toll-like receptors; MD2, myeloid differentiation factor 2.

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has examined the antitumor effect of *A. membranaceus* in humans and animals [7]. There are 24 polysaccharides extracted from the dried root of *A. membranaceus* defined by monosaccharide composition, molecular weight and position of glycosidic linkages, etc. [8,9]. The immunomodulatory and antitumor effects of *A. membranaceus* polysaccharide (APS) have been attributed to its structural features [9,10]. APS has the molecular weight range of 8.7–4800 kDa comprised as various monosaccharide composition [9]. The chemical structure of APS with immunomodulatory properties, in particular, was an α -(1 → 4)-linked dextran backbone with α -(1 → 6) linked branches every 10 residues and molecular weight was approximately 21 kDa [11].

Melanoma, which is a highly aggressive cancer [12,13], spreads from primary sites, such as skin to the lymphatic system through the blood vessels and metastasizes to multiple organs, including the lymph node (LN), lung, liver, and brain [12–14]. The lung is the most frequent site for melanoma metastasis [15,16]. Patients with lung metastasis of melanoma are known to have poor clinical outcomes and overall survival [15–17]. The advances in chemoradiotherapy have resulted in limited

success in improving the clinical outcomes of patients with melanoma [18]. Additionally, the development of immunotherapeutic strategies, such as antitumor vaccines and immune checkpoint inhibitors has improved the treatment outcomes of patients with metastatic melanoma [19,20]. The therapeutic strategy for patients with metastatic melanoma exhibiting upregulated expression of programmed cell death-ligand 1 (PD-L1) is the administration of monoclonal antibodies (mAbs) against programmed cell death-1 (PD-1) [19,20]. PD-1 negatively regulates immune responses mediated by the B cells, dendritic cells (DCs), natural killer (NK) cells and T cells [21,22]. PD-L1 expressed in the cancer cells binds to PD-1 on the immune cells, which results in the inhibition of immune responses against cancer [21,22]. The administration of anti-PD-1/PD-L1 mAbs has improved the clinical outcomes of patients with cancer. However, the combination of mAb therapy and other therapeutic approaches, such as adjuvant therapy can aid to improve the anti-cancer efficacy of anti-PD-1/PD-L1 mAbs [23,24].

Cell-mediated immunity against cancer is induced by complex interactions between antigen-presenting cells (APCs) and cytokines, thereby resulting in cancer cell death [25]. Cytotoxic T-lymphocytes (CTLs) and NK cells are the major immune cells involving cell-mediated immune responses, especially those against tumors [26–28]. APCs, such as macrophages and DCs activate CTLs [29]. Activated DCs express costimulatory molecules and promote the differentiation and activation of CD8 T cells to CTLs [25,30]. Additionally, DCs regulate the activation of NK cells through the secretion of interleukin (IL)-12. However, some stimuli can directly promote NK cell activation [31–33]. Activated CTL and NK cells produce pro-inflammatory cytokines and cytotoxic mediators, which are the main mediators that induce target cell death [34–36].

Previous studies have demonstrated that APS activated mouse bone marrow-derived DCs and macrophages, as well as human monocyte-derived DCs [37–39]. Moreover, APS promoted proliferation and antibody production in the splenic B cells [2]. Furthermore, APS are reported to activate the immune responses against cancer in mice [38,40]. Although the immunostimulatory effect of APS has been previously examined, the effect of intranasal administration of APS on the activation of mucosal immune cells, as well as on the anti-cancer effects of anti-PD-L1 mAb, have not been examined. This study aimed to examine the effect of intranasal administration of APS on the activation of immune cells in the mediastinal LN (mLN), as well as on the anti-cancer effects of anti-PD-L1 mAb, in mice.

2. Materials and methods

2.1. Reagents and monosaccharide analysis

APS were purchased from Shanghai Yuan Ye Biotechnology Co., Ltd. (Shanghai, China). APS was treated with 4 M trifluoroacetic acid at 100 °C for 6 h. The hydrolyzed APS was reduced with sodium borodeuteride, followed acetylated by acetic anhydride. The weight-average molecular weight of APS was analyzed using gel permeation chromatography (GPC), combined with Agilent Technologies 1200 series high-performance liquid chromatography (Agilent Technologies, CA, USA). The weight-average molecular weight of APS was calculated by GPC software, and the homogeneity was determined by polydispersity index (PDI). The product injected into the gas chromatograph/mass spectrometer (GC/MS, GCMS-QP2010Ultra, Shimadzu, Japan).

2.2. Fourier transform infrared (FT-IR) spectrometry analysis

FT-IR spectrum of APS was recorded between 500 and 3500 cm^{-1} using a Perkin Elmer Spectrum100 FT-IR spectrophotometer (Perkin Elmer Inc., Wellesley, MA) at Core Research Support Center for Natural Products and Medical Materials in Yeungnam University.

2.3. Mice and cell lines

C57BL/6 mice (aged 6–10 weeks), which were given from the Shanghai Public Health Clinical Center (SPHCC), were maintained under specific pathogen-free conditions at the Animal Center of SPHCC. The mice were housed in a room at 20–22 °C with 50%–60% humidity and fed standard rodent chow and water. The study was performed following the guidelines of the Institutional Animal Care and Use Committee at the SPHCC. Using of mice in the study was approved by the Ethics of Animal Experiments Committee of the SPHCC (Approval number: 2018-A049–01). Mice were euthanized through CO₂ inhalation to minimize pain to the animals. B16F10 (murine melanoma cell line; ATCC, CRL-6475) cells were cultured in RPMI-1640 medium supplemented with 100 $\mu\text{g}/\text{mL}$ streptomycin, 1 M hydroxyethyl piperazineethanesulfonic acid, 10% fetal bovine serum, 2 mM glutamine, and 100 U/mL penicillin in a humid incubator at 5% CO₂ and 37 °C.

2.4. Reagent and antibodies

Lipopolysaccharide (LPS; O111:B4) was purchased from Sigma-Aldrich (St. Louis, MO, USA). Isotype control antibodies (IgG1, IgG2a, and IgG2b) and anti-CD3 (17A2), anti-CD11c (N418), anti-CD8 α (53–6.7), anti-CD40 (3/23), anti-CD69 (3/23), anti-CD80 (16-10A1), anti-CD86 (GL-1), anti-NK1.1 (PK136), anti-IL-12p70 (C18.2), anti-perforin (GL-1), anti-granzyme B (C18.2), and anti-interferon (IFN)- γ (MP5-20F3) antibodies were purchased from BioLegend (CA, USA). Anti-major histocompatibility complex (MHC) class I (28–8–6), anti-MHC class II (M5/114.15.2), and anti-tumor necrosis factor- α (TNF- α) (MP6-XT22) antibodies were obtained from eBioscience (CA, USA).

2.5. Flow cytometric analysis

The cells were incubated with control antibodies (unlabeled isotypes) to inhibit non-specific binding. Next, the cells were washed and stained with fluorescence probe-conjugated antibodies on ice for 30 min, followed by washing to remove the unbound antibodies. The cells were then analyzed using a Novocyte flow cytometer and NovoExpress® software from (ACEA Biosciences Inc., CA, USA). Dead cells stained with 4',6-diamidino-2-phenylindole (DAPI, Sigma-Aldrich, MO, US) were excluded from the analysis.

2.6. Analysis of mouse mLN DCs

C57BL/6 mice were intranasal administration of phosphate-buffered saline (PBS), APS (50 mg/kg bodyweight) or LPS (0.2 mg/kg bodyweight). Eighteen hours after treatment, the mice were sacrificed and mLN was harvested as described previous study [41,42]. Briefly, the mice lay with the stomach up and cut the ribs with scissors. The lungs was overlapped in one direction. The mLN was harvested in air way by forceps. The mLN was ground using two cover glasses and filtered through a nylon mesh (pore size: 100 nm). The cells were stained with anti-CD3, anti-CD90.1, anti-B220, anti-Gr-1, anti-CD49b, anti-TER-119, and anti-CD11c antibodies. DCs in mLN, which were defined as lineage[−]CD11c⁺ cells, were analyzed using a flow cytometer (Novocyte, CA, USA).

2.7. Analysis of NK and T cells in the mLN

C57BL/6 mice were intranasally administered with PBS, 50 mg/kg APS or 0.2 mg/kg LPS in two consecutive treatments at two-day interval. On day 2 post-final treatment, mLN was harvested and a single cell suspension was prepared. The cells were then treated with 1 μM ionomycin and 50 ng/mL phorbol 12-myristate 13-acetate (both from Sigma-Aldrich) for 4 h with the addition of monensin

solution (BioLegend, CA, USA) in the final 2 h. The cells were stained with anti-CD3, anti-CD4, anti-CD8 and anti-NK1.1 at 4 °C for 20 min. The cells further stained with intracellular antibodies after fixation and permeabilization. The CD3⁺NK1.1⁺ or CD3⁺NK1.1[−] cells were defined as NK or T cells, respectively.

2.8. Intracellular cytokine staining

The cells were stained with antibodies against surface proteins, fixed, and permeabilized using Cytofix/Cytoperm buffer (eBioscience, CA, USA). Next, the cells were washed with Perm/Wash buffer from (eBioscience, CA, USA) and incubated with antibodies against cytokines in Perm/Wash buffer for 30 min at room temperature. The staining was inhibited using the anti-Fc antibody. Isotype control IgGs were used as negative controls in all experiments.

2.9. Enzyme-linked immunosorbent assay (ELISA)

Bronchoalveolar lavage (BAL) fluid was harvested from the mice. The concentrations of interleukin (IL)-1 β , IL-6, IL-12p40, IFN- γ , and TNF- α in the BAL fluid were measured 18 h after treatment of PBS, APS or LPS in triplicates using ELISA kits (BioLegend, CA, USA). Concentration of IFN- γ (BioLegend), TNF- α , Perforin (Abbkine, Wuhan, China), and Granzyme B (LSBIO, Seoul, Korea) in BAL fluid were measured after the second APS treatment by ELISA kit.

2.10. Tumor treatment

C57BL/6 mice were intravenously injected with B16 melanoma cells (0.5×10^6 /100 μ L). The mice were divided 4 groups; PBS, anti-PD-L1 antibody (10 mg/kg bodyweight), APS (50 mg/kg bodyweight) and the combination of anti-PD-L1 antibody and APS. The PBS and APS were administered by intranasal injection to the mice, while the anti-PD-L1 antibody was treated by intraperitoneal injection. On day 5 post-tumor cell injection, the mice received treatment for every two days. The mice were monitored survival and measured changes of body weight during treatment. Some mice were euthanized on day 14 post-B16 tumor cell injection. The lung tissue was harvested to analyze the infiltration of B16 melanoma cells.

2.11. Hematoxylin and eosin (H&E) staining

H&E staining was performed as described previously [30,43–46]. Briefly, the lung samples were fixed in 4% formaldehyde, embedded in paraffin, and sectioned at a thickness of 5- μ m. The sections were de-paraffinized, rehydrated, stained with H&E, and scored for tissue damage.

2.12. Depletion of NK cells and CD8 T cells

On day 4 post-B16 tumor cell injection, the mice were separated with 4 groups; Isotype control IgG, anti-NK1.1 antibody (50 μ g/100 μ L, PK136), anti-CD8 antibody (50 μ g/100 μ L, YTS169.4 L), and combination of anti-NK1.1 and anti-CD8 antibodies (both from Bioxcell, NH, USA) were administered intraperitoneally at two-day interval. The mice were treated with combination of APS (50 mg/kg bodyweight) and anti-PD-L1 antibody (10 mg/kg bodyweight) on day 5 post-tumor cell injection. The survival of the mice was monitored until day 30 post-tumor injection.

2.13. Statistical analysis

All data are expressed as mean $\pm$ standard error of mean. The data were analyzed using one-way or two-way analysis of variance, followed by Tukey's multiple comparison test and Mann-Whitney *U* test in SPSS

(IBM, Armonk, NY, USA). The differences were considered significant at $p < 0.05$.

3. Results

3.1. Intranasal administration of APS promoted DC migration to the mLN

As the molecular weight and chemical composition of APS were the among the important information of bioactivity, the molecular weight and homogeneity of APS were established by GPC. Table 1 and Fig. S1a showed that the weight average molecular weight of APS revealed approximately 17.38 kDa and homogeneity was 1.738 by polydispersity index of APS. Moreover, to determine the monosaccharide composition in APS, hydrolyzed APS was analyzed by GC/MS as shown in Table 2 and Fig. S1b. The results revealed that the main content of APS was glucose (95.0%), along with the minor amount of arabinose (2.9%), xylose (0.7%), mannose (0.7%) and galactose (0.6%), which the contains of monosaccharide were similar with previous studies [43,47]. As shown in FT-IR spectrum, broad band at 3290 cm^{-1} related to the hydroxyl group of O—H stretching vibration (Fig. 1a) [48]. Weak band at 2928 cm^{-1} was demonstrated to the C—H stretching vibration [49]. The peak at 1604 and 1407 cm^{-1} was caused by C=O asymmetric and symmetric stretching vibration [43]. The strong absorption peak at 1018 cm^{-1} designated as the stretching vibration of C—O—C and glycosidic bond [50]. The peak at 859 and 759 cm^{-1} indicated the presence of β -linked D-mannopyranose units and α -linked D-galactopyranose [50].

3.2. Intranasal administration of APS promoted DC migration to the mLN

Next, to evaluate the effect of APS on mucosal immune activation, C57BL/6 mice were intranasally administered with 50 mg/kg of APS (Fig. 1b). The concentration of APS was determined by referring to previous studies [51–54]. The population and number of DCs in the mLN were examined 18 h post-treatment described in the materials and methods. As shown in Fig. 1c, DCs in the mLN were identified based on the lineage[−]CD11c⁺ cells in live leucocytes. The proportion of lineage[−]CD11c⁺ DCs in the mLN of the APS-treated group was higher than that in the mLN of the PBS-treated group (Fig. 1d). Similarly, the absolute number of DCs in the mLN of the APS-treated group was significantly higher than that in the mLN of the PBS-treated group (Fig. 1e). Compared with that in the PBS-treated group, the expression of CC-chemokine receptor 7 (CCR7) a marker of DCs that migrate into the LN after maturation, was markedly upregulated in the APS-treated group (Fig. 1f). The migration of DCs into the mLN was similar between the APS-treated and LPS-treated groups. This indicated that the intranasal administration of APS can promote the migration of DCs into mLN.

3.3. APS activated DCs in the mLN

Next, the effect of APS on mLN DC activation was examined. The mLN was harvested 18 h post- PBS, APS or LPS treatment and the expression levels of co-stimulatory molecules and MHC classes I and II in the DCs were measured. Compared with those in the PBS-treated group, the expression levels of CD40, CD80, CD86, MHC class I, and MHC class II were markedly upregulated in the APS-treated group (Fig. 2a and b). Additionally, the levels of pro-inflammatory cytokines (IL-6, IL-12, and TNF- α) in BAL fluid were much higher in the APS-treated group than those in the PBS-treated group (Fig. 2c). The

Table 1
Weight-average molecular weight (Mw) and polydispersity index (PDI) of APS.

Mw	PDI
17.39 kDa	1.738

Table 2
Monosaccharide composition of APS.

Monosaccharide composition	Content (%)
Glucose	95.0 ± 1.2
Arabinose	2.9 ± 0.7
Xylose	0.7 ± 0.5
Mannose	0.7 ± 0.1
Galactose	0.6 ± 0
Rhamnose	N.D. ^a
Fucose	N.D.

^a Abbreviation: N.D., not detected.

expression levels of activation markers in DCs were similar between the APS-treated and LPS-treated groups. In contrast, the levels of the pro-inflammatory cytokines in the BAL fluid in the APS-treated group were markedly lower than those in the LPS-treated group (Fig. 2c). Additionally, APS did not upregulate the levels of IL-1 β in BAL fluid, whereas LPS upregulated IL-1 β in the BAL fluid (Fig. 2c). These data suggest that intranasal administration of APS can promote the activation of DCs in the mLN and the production of pro-inflammatory cytokines in the lungs.

3.4. APS induced NK cell and T cell activation in mLN

NK and T cells are majorly activated by DCs [25]. We next examined whether the intranasal administration of APS can induced NK and T cell activation in mLN. C57BL/6 mice were intranasally administered with PBS, APS (50 mg/kg), or LPS (0.2 mg/kg). The mice were again administered with the same concentration of PBS, APS, or LPS two days later. The mLN was harvested on day 2 post-last treatment and the activation of NK cells and T cell cells was analyzed. Treatment with APS markedly increased the number of CD3⁺NK1.1⁺ cells, which defined as NK cells (Fig. 3a). Compared with that in the mLN of the PBS-treated group, the number of NK cells was significantly higher in the mLN of the APS-treated group (Fig. 3b). The percentage and number of T cells in the mLN were similar between the APS-treated and PBS-treated groups (Fig. 3a and c). We also measured activation of the NK cells by APS and found that treatment with APS markedly upregulated the intracellular producing levels of IFN- γ , granzyme B, and perforin in the mLN NK cells (Fig. 3d). Additionally, CD4 and CD8 T cells produced IFN- γ levels in mLN (Fig. 3e and f) and lung (Fig. S2) were also significantly upregulated in the APS-treated group compared to PBS treatment (Fig. 3e and f). However, APS did not promote the production of TNF- α in the CD4, although it has considerably increased in CD8 T cells

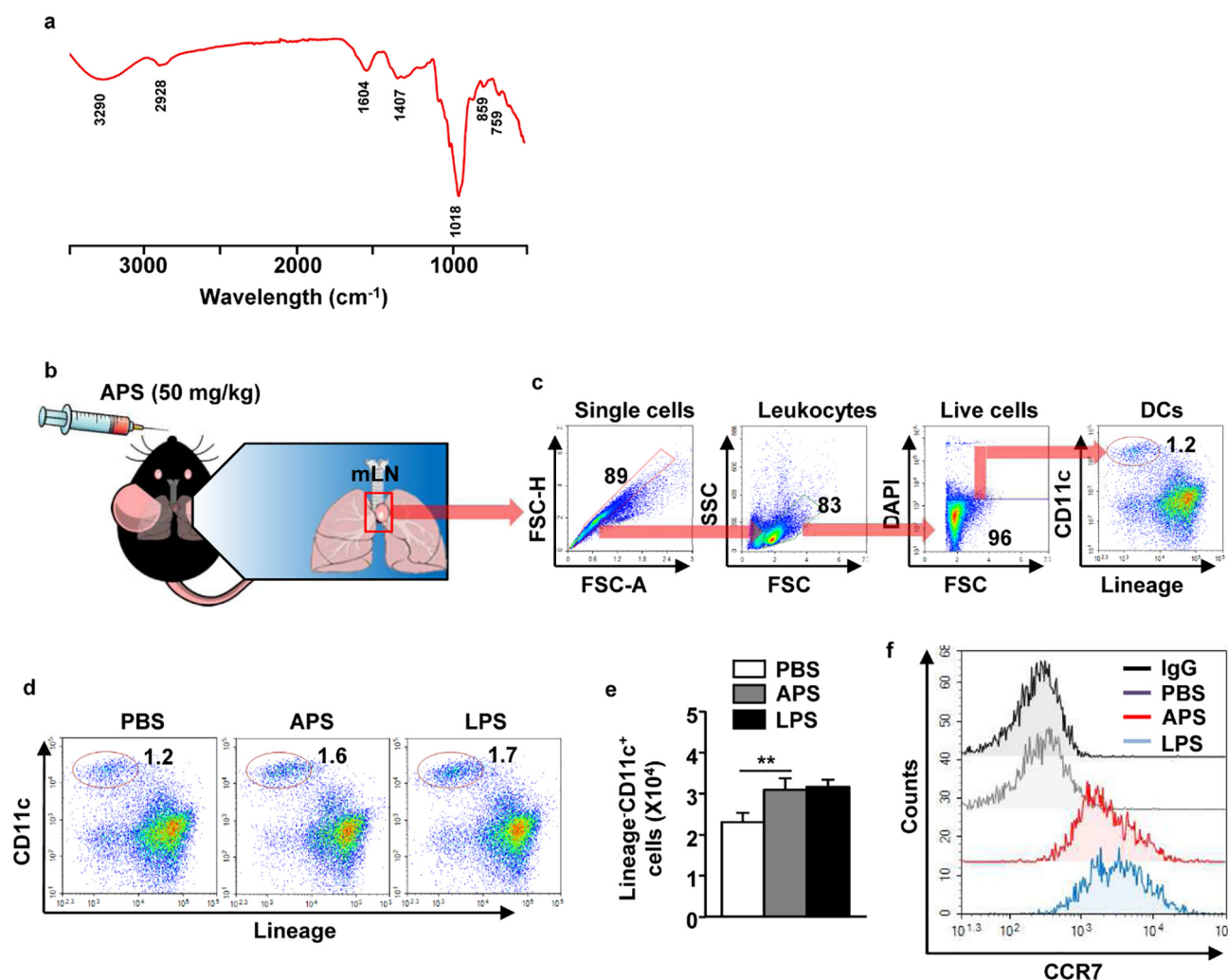


Fig. 1. *Astragalus membranaceus* polysaccharides (APS) induced migration of dendritic cells (DCs) to mediastinal lymph node (mLN). (A) FT-IR spectrum of APS. (B) Illustration of intranasal administration with APS. C57BL/6 mice were intranasally administered with 50 mg/kg body weight of APS. (C) Strategy of flow cytometry analysis for mLN DCs was shown. (D) At 18 h post-treatment, mLN was harvested and the DCs in mLN were analyzed using flow cytometry. Dot plots showing the population of DCs in mLN. (E) The number of lineage⁺CD11c⁺ DCs in mLN. (F) The expression levels of CC-chemokine receptor 7 (CCR7) in the mLN DCs. ($n = 6$ mice, one-way ANOVA, mean $\pm$ SEM, ** $p < 0.01$).

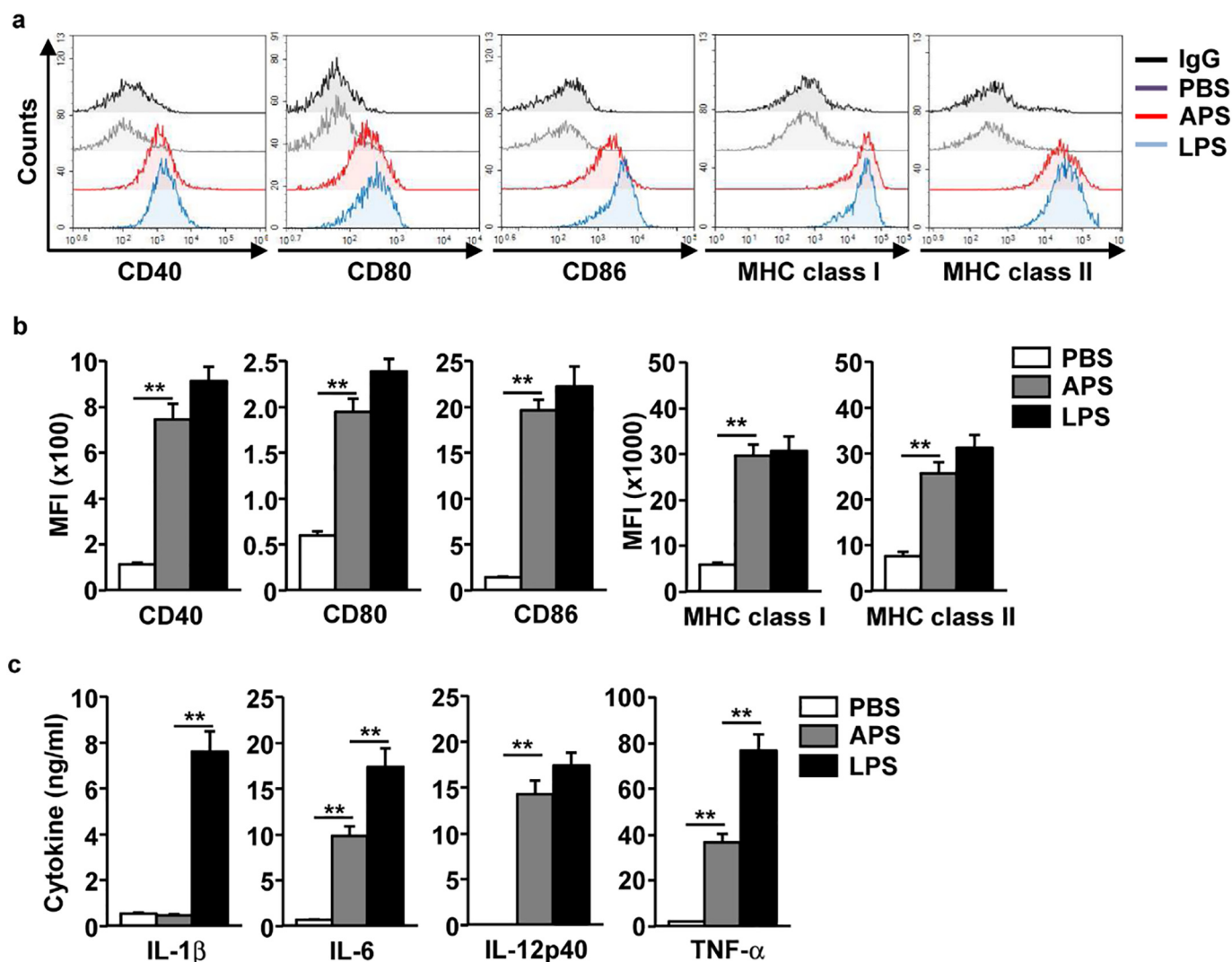


Fig. 2. Intranasal administration of APS promoted the activation of mLN DCs in mice. C57BL/6 mice were treated with APS and the mLN was harvested as indicated in the legend of Fig. 1. (A) Histogram showing the expression levels of the indicated surface markers in mLN DCs. (B) Mean fluorescence of intensity (MFI) of the surface markers in mLN DCs. (C) The concentrations of interleukin (IL)-1 β , IL-6, IL-12p40, and tumor necrosis factor (TNF)- α in the bronchoalveolar (BAL) fluid were measured using enzyme-linked immunosorbent assay (ELISA). (n = 6 mice, one-way ANOVA, mean $\pm$ SEM, **p < 0.01).

(Fig. 3e and f). Compared with those in the BAL fluid of the PBS-treated group, the levels of these cytokines and cytotoxic mediators were markedly upregulated in the BAL fluid of the APS-treated group (Fig. 3g). Thus, these findings suggest that intranasal administration of APS can promote NK and T cell-mediated immune activation in the lungs.

3.5. Intranasal administration of APS potentiates the anti-cancer effects of anti-PD-L1 antibodies against pulmonary metastatic melanoma

Since intranasal administration of APS promoted immune activity in lung and mLN, we next examined whether APS enhances anti-cancer effect of anti-PD-L1 antibodies against pulmonary metastatic melanoma was examined. Treatment with APS or anti-PD-L1 antibodies prolonged the survival of mice injected with B16 melanoma cells compared to PBS-treated controls (Fig. 4a). Interestingly, mice treated with the combination of APS and anti-PD-L1 antibodies survived even on day 30 post-tumor injection (Fig. 4a). Moreover, this treatment combination did not affect the alteration of bodyweight of mice (Fig. 4b). In addition, pulmonary infiltration of B16 cells was markedly inhibited upon treatment with the combination of APS and anti-PD-L1 antibody (Fig. 4c).

To evaluate the role of immune cells in the anti-cancer effects of the APS/anti-PD-L1 antibody combination, the NK and CD8 T cells were

depleted in mice (Fig. S3) injected with metastatic melanoma cells during treatment. The depletion of NK or T cells mitigated the APS/anti-PD-L1 Ab combination-induced anti-cancer effect as evidenced by the death of mice within day 18 of tumor injection (Fig. 4d). These data demonstrated that intranasal administration with APS can potentiate the anti-cancer effect of anti-PD-L1 antibodies against pulmonary metastatic melanoma.

4. Discussion

Natural polysaccharides are reported to exhibit immunostimulatory effects in humans and mice [38,40,44–46,55,56]. *A. membranaceus* is a traditional and safe herbal medicine that is widely used in Korea and China owing to its low toxicity [1–5]. Previous studies have demonstrated that APS exerted immunostimulatory effects on DCs and lymphocytes [3,37,38]. Most studies on the immunostimulatory function of APS were performed using an *in vitro* culture system, which does not mimic the *in vivo* immune responses. This study demonstrated that intranasal administration of APS exerts immunostimulatory effects in the mLN and lungs of mice. APS promoted the activation of DCs, NK cells, and T cells in the mLN (Figs. 2 and 3). The APS-induced activation of NK and T cells resulted in the enhanced production of pro-

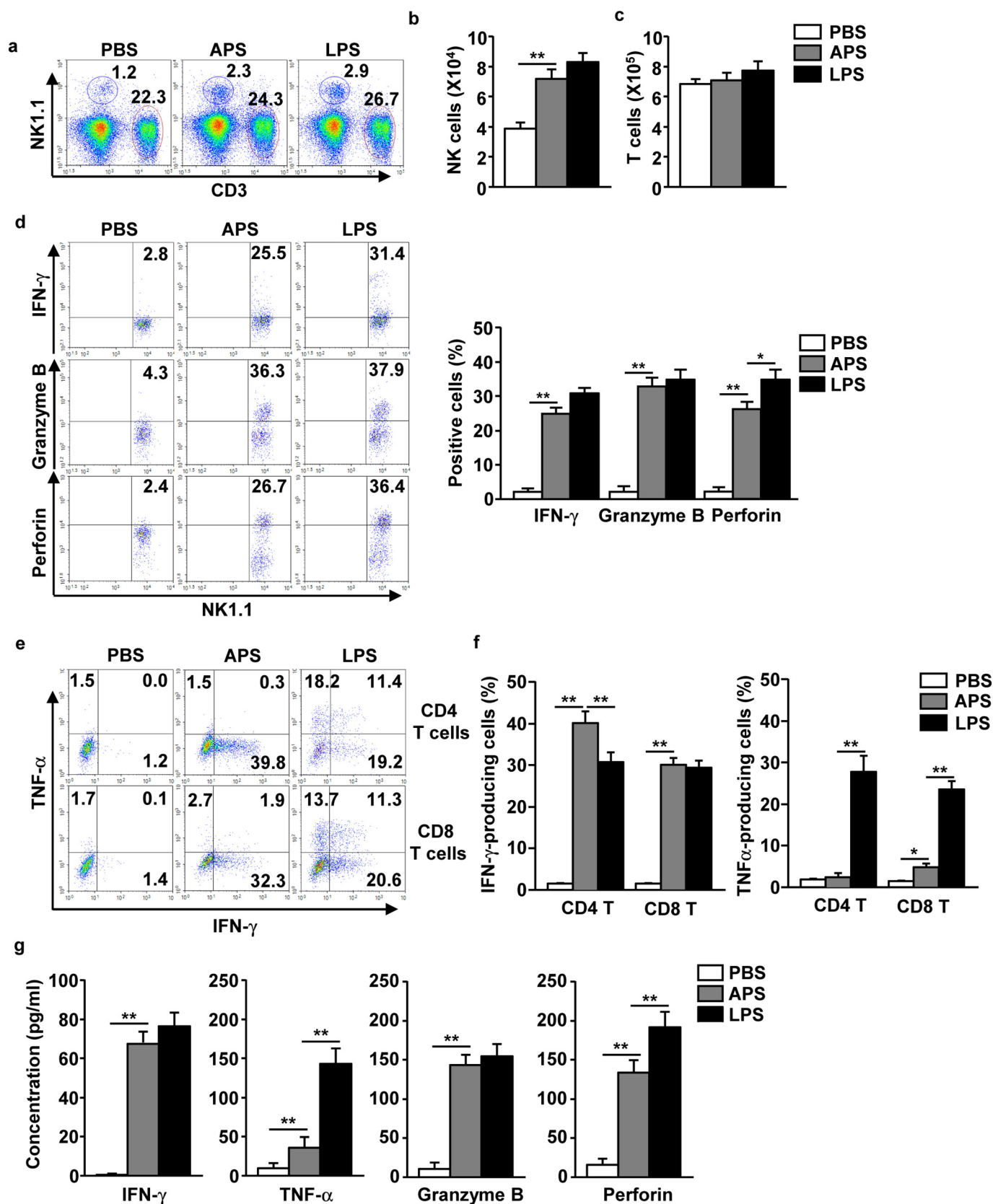


Fig. 3. APS activated the natural killer (NK) cells and T cells in the mLN. Time-dependent effects of APS on splenic DC activation. C57BL/6 mice were intranasally administered with PBS, APS (50 mg/kg), or lipopolysaccharide (LPS, 0.2 mg/kg) twice with a two-day interval. (A) Frequency and proportion of NK and T cells in mLN. (B) Number of NK cells and (C) T cells in the mLN. (D) The levels of interferon (IFN)- γ , granzyme B, and perforin in the mLN NK cells were measured using flow cytometry (left panel). Mean percentage of cells exhibiting the expression of IFN- γ , granzyme B, and perforin. (E) The levels of IFN- γ , and TNF- α in CD4 or CD8 T cells. (F) Mean percentage of cells exhibiting the expression of IFN- γ (left panel) and TNF- α (right panel) in the CD4 and CD8 T cells. (G) Concentrations of IFN- γ , TNF- α , granzyme B, and perforin in the BAL fluid were measured using ELISA. (n = 6 mice, one-way ANOVA, mean $\pm$ SEM, **p < 0.01).

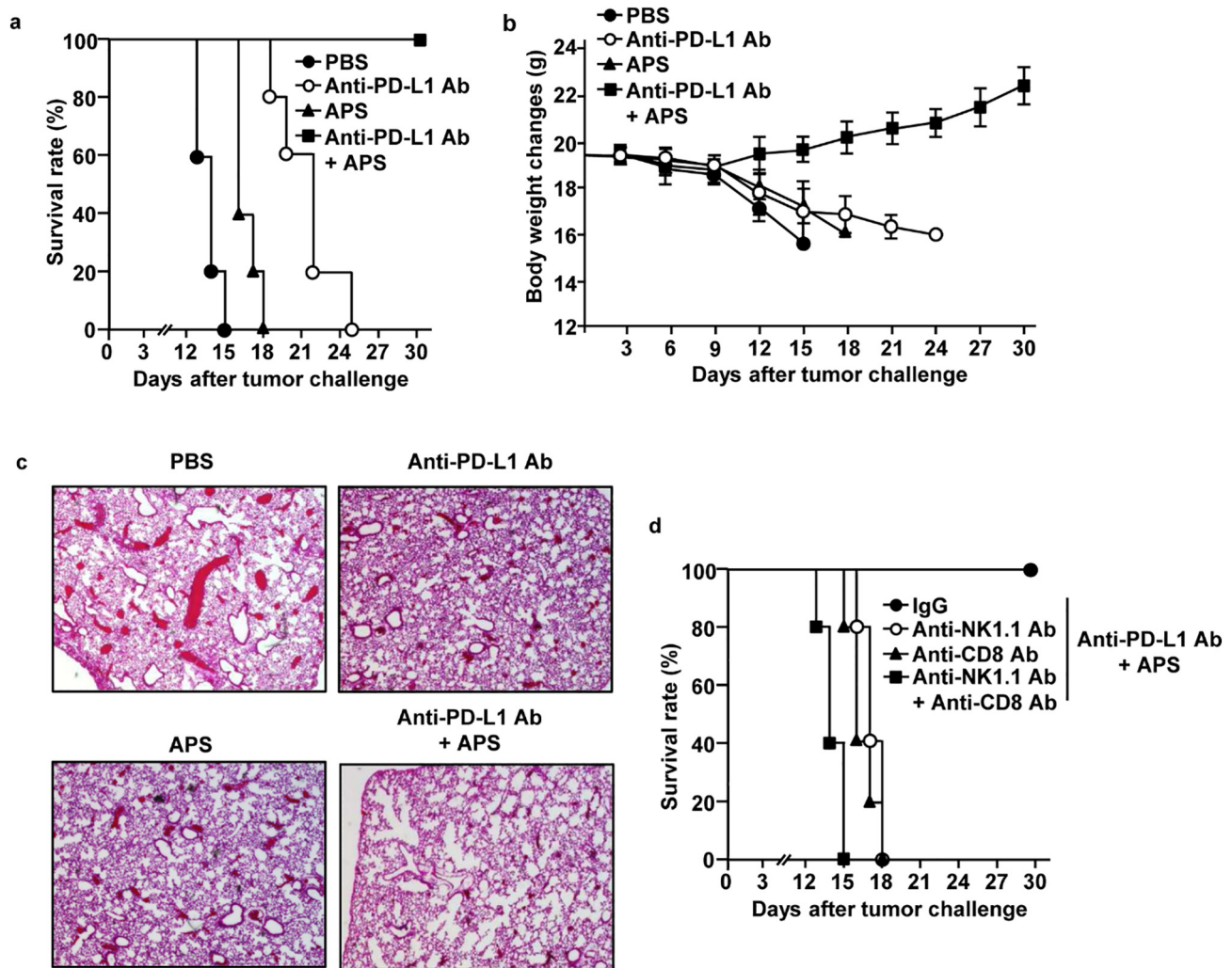


Fig. 4. Intranasal administration of APS potentiates the anti-cancer effects of anti-programmed death-ligand 1 (PD-L1) antibody (Ab) in mice. C57BL/6 mice were intravenously injected with B16 melanoma cells ($0.5 \times 10^6/100 \mu\text{L}$). On day 5 post-cancer cell injection, mice were intranasally administered with APS (50 mg/kg), intraperitoneally administered with anti-PD-L1 Ab (10 mg/kg), or a combination of APS and anti-PD-L1 Ab at two-day intervals. Survival rate (A) ($n = 5$) and bodyweight (B) ($n = 5$) of mice belonging to different treatment groups. (C) Analysis of pulmonary infiltration of melanoma cells using hematoxylin and eosin staining on day 14 post-melanoma cell injection. (D) Survival rate of NK cell-depleted or CD8 cell-depleted mice ($n = 5$).

inflammatory cytokines and cytotoxic mediators in BAL fluid, which mediate cancer cell death (Fig. 3). Thus, APS potentiated the growth-inhibitory activity of anti-PD-L1 mAbs against metastatic melanoma in mice (Fig. 4). These findings demonstrated that APS can activate mucosal immune cells and that it can function as an adjuvant to potentiate the anti-cancer activity of immune checkpoint inhibitors.

Fucoidan, a polysaccharide extracted from brown agar [9,57], contains fucose as the main monosaccharide [58]. Various studies have reported the biological activities of fucose-containing polysaccharides, such as immunostimulatory activities in mice and humans [46,55,57,59,60]. APS comprise arabinose, galactose, glucose, and xylose [9]. The concentration of fucose in APS is markedly lower than that in fucoidan or ascopyllan, whereas glucose concentration in APS is much higher than those polysaccharide [9]. Therefore, immune stimulatory effect of APS may be attributed to glucose or other components that are generated upon degradation of APS. In future, we will examine the effect of monosaccharides in APS on the activation of immune cells.

DCs are APCs that promote adaptive immune activation [44,61,62]. The activation of DCs is mediated by the stimulation of pattern recognition receptors, such as toll-like receptors (TLRs) and scavenger receptors [30,44]. The stimulation of TLRs by their ligand promotes the migration of DCs to the LN or spleen, upregulates the expression of co-

stimulatory molecules, and enhances the secretion of pro-inflammatory cytokines [25,63]. Activated DCs promote the differentiation and activation of T cells [25]. TLR4 is the most studied stimulatory receptor in DCs [30,44]. LPS, which is the ligand of TLR4, promotes the activation of DCs [44]. APS induced RAW264.7 cells, a murine macrophage cell line [5], through the TLR4 signaling pathway. However, the patterns of immune cell activation and cytokine production in the mLN and lung in the APS-treated group were markedly different from those in the LPS-treated group. APS did not enhance the production of IL-1 β in the lung (Fig. 2c). In contrast, treatment with LPS markedly upregulated the levels of IL-1 β (Fig. 2c). Additionally, the levels of TNF- α in the APS-treated group were markedly different from those in the LPS-treated group (Fig. 2c). Treatment with APS did not affect the bodyweight of mice (Fig. 4b). In contrast, treatment with LPS gradually decreased the bodyweight of mice (Fig. 4b). The differential effects APS and LPS on immune cell activation may be attributed to the activation of different signaling pathways. LPS activates DCs through TLR4 and myeloid differentiation factor 2 (MD2) [30,43–45,64]. MD2, which is a link between LPS and TLR4, activates the signaling pathway in DCs [64]. TLR4/MD2 stimulation mainly induces toxicity and IL-1 β production in mice [64]. In this study, APS did not induce IL-1 β production and toxicity in mice. Hence, APS may stimulate TLR4 directly without MD2. To determine the stimulatory receptor of APS, further studies are required using TLR4 and MD2 knockout mice.

Immunotherapy is a promising strategy for treating cancer, especially metastasizing cancer [20,23]. Moreover, immune checkpoint inhibitors are one of the standard treatments for patients with metastatic melanoma [20]. The activation of anti-cancer immunity by immune checkpoint inhibitors does not completely inhibit the progression of metastatic melanoma [23]. To address this issue, various studies have examined the anti-cancer efficacy of the combination of immune checkpoint inhibitors and adjuvants in animals and humans [65,66]. In this study, a novel immune stimulator was demonstrated to potentiate the anti-cancer effect of immune checkpoint inhibitors (Fig. 4). Compared with other adjuvants, APS can potentially stimulate immune cell activation. The intranasal administration of APS activated the immune cells in the mLN. Additionally, APS exhibited lower toxicity than LPS in mice as described in the result section. Therefore, APS are potential therapeutic agents for improving the anti-cancer efficacy of immune checkpoint inhibitors.

5. Conclusion

The intranasal administration of APS activated DCs, CTLs, and NK cells in the mLN, which resulted in the potentiation of growth-inhibitory activity of anti-PD-L1 mAbs against pulmonary metastatic melanoma. The findings of this study demonstrated that APS can be used as a mucosal adjuvant for enhancing anti-cancer immunity in mice.

CRedit authorship contribution statement

Juyoung Hwang: Conceptualization, Formal analysis, Writing-Original Draft **Wei Zhang:** Conceptualization, Data Curation, Writing-Original Draft **Yadav Dhananjay:** Data Curation **Eun-Koung An:** Data Curation **Minseok Kwak:** Supervision **SangGuan You:** Supervision, Investigation **Peter Chang-Whan Lee:** Supervision **Jun-O Jin:** Conceptualization, Writing- Reviewing and Editing.

Declaration of competing interest

The authors have declared no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2021.05.073>.

References

- [1] Z. Zheng, D. Liu, C. Song, C. Cheng, Z. Hu, Studies on chemical constituents and immunological function activity of hairy root of *Astragalus membranaceus*, *Chin. J. Biotechnol.* 14 (2) (1998) 93–97.
- [2] B.M. Shao, W. Xu, H. Dai, P. Tu, Z. Li, X.M. Gao, A study on the immune receptors for polysaccharides from the roots of *Astragalus membranaceus*, a Chinese medicinal herb, *Biochem. Biophys. Res. Commun.* 320 (4) (2004) 1103–1111.
- [3] X. Du, B. Zhao, J. Li, X. Cao, M. Diao, H. Feng, X. Chen, Z. Chen, X. Zeng, *Astragalus* polysaccharides enhance immune responses of HBV DNA vaccination via promoting the dendritic cell maturation and suppressing Treg frequency in mice, *Int. Immunopharmacol.* 14 (4) (2012) 463–470.
- [4] S.M. Chen, Y.S. Tsai, S.W. Lee, Y.H. Liu, S.K. Liao, W.W. Chang, P.J. Tsai, *Astragalus membranaceus* modulates Th1/2 immune balance and activates PPAR γ in a murine asthma model, *Biochem. Cell Biol.* 92 (5) (2014) 397–405.
- [5] Y. Tian, X. Li, H. Li, Q. Lu, G. Sun, H. Chen, *Astragalus mongolicus* regulate the toll-like-receptor 4 mediated signal transduction of dendritic cells to restrain stomach cancer cells, *Afr. J. Tradit. Complement. Altern. Med.* 11 (3) (2014) 92–96.
- [6] T.S. Yeh, H.L. Chuang, W.C. Huang, Y.M. Chen, C.C. Huang, M.C. Hsu, *Astragalus membranaceus* improves exercise performance and ameliorates exercise-induced fatigue in trained mice, *Molecules* 19 (3) (2014) 2793–2807.
- [7] K.K. Auyeung, Q.B. Han, J.K. Ko, *Astragalus membranaceus*: a review of its protection against inflammation and gastrointestinal cancers, *Am. J. Chin. Med.* 44 (1) (2016) 1–22.
- [8] M. Jin, K. Zhao, Q. Huang, C. Xu, P. Shang, Isolation, structure and bioactivities of the polysaccharides from *Angelica sinensis* (Oliv.) Diels: a review, *Carbohydr. Polym.* 89 (3) (2012) 713–722.
- [9] M. Jin, K. Zhao, Q. Huang, P. Shang, Structural features and biological activities of the polysaccharides from *Astragalus membranaceus*, *Int. J. Biol. Macromol.* 64 (2014) 257–266.
- [10] A.J. Liu, J. Yu, H.Y. Ji, H.C. Zhang, Y. Zhang, H.P. Liu, Extraction of a novel cold-water-soluble polysaccharide from *Astragalus membranaceus* and its antitumor and immunological activities, *Molecules* 23 (1) (2017).
- [11] Y. Niu, H. Wang, Z. Xie, M. Whent, X. Gao, X. Zhang, S. Zou, W. Yao, L. Yu, Structural analysis and bioactivity of a polysaccharide from the roots of *Astragalus membranaceus* (Fisch) Bge. var. *mongolicus* (Bge.) Hsiao, *Food Chem.* 128 (3) (2011) 620–626.
- [12] E. Atallah, L. Flaherty, Treatment of metastatic malignant melanoma, *Curr. Treat. Options in Oncol.* 6 (3) (2005) 185–193.
- [13] K.V. Liang, S.O. Sanderson, G.S. Nowakowski, A.S. Arora, Metastatic malignant melanoma of the gastrointestinal tract, *Mayo Clin. Proc.* 81 (4) (2006) 511–516.
- [14] A. Caldaria, R. Giuffrida, N. di Meo, L. Massari, C. Dianzani, S.P. Cannavo, F. Degrossi, E. Casablanca, I. Zalaudek, C. Conforti, Diagnosis and treatment of melanoma bone metastasis: a multidisciplinary approach, *Dermatol. Ther.* 33 (6) (2007), e14193.
- [15] P.P. Velu, C. Cao, T.D. Yan, Current surgical management of melanoma metastases to the lung, *J. Thorac. Dis.* 5 (Suppl. 3) (2013) S274–S276.
- [16] F. Tas, Metastatic behavior in melanoma: timing, pattern, survival, and influencing factors, *J. Oncol.* 2012 (2012), 647684.
- [17] C. Kyriakopoulos, G. Zarkavelis, A. Andrianopoulou, A. Papoudou-Bai, D. Stefanou, S. Boussios, G. Penthroudis, Primary pulmonary malignant melanoma: report of an important entity and literature review, *Case Rep. Oncol. Med.* 2017 (2017), 8654326.
- [18] R. Luis, C. Brito, M. Pojo, Melanoma metabolism: cell survival and resistance to therapy, *Adv. Exp. Med. Biol.* 1219 (2020) 203–223.
- [19] J. Han, X. Xu, Z. Liu, Z. Li, Y. Wu, D. Zuo, Recent advances of molecular mechanisms of regulating PD-L1 expression in melanoma, *Int. Immunopharmacol.* 88 (2020), 106971.
- [20] A.A. Albittar, O. Alhalabi, I.C. Glitza Oliva, Immunotherapy for melanoma, *Adv. Exp. Med. Biol.* 1244 (2020) 51–68.
- [21] H. Hayashi, K. Nakagawa, Combination therapy with PD-1 or PD-L1 inhibitors for cancer, *Int. J. Clin. Oncol.* 25 (5) (2020) 818–830.
- [22] D.H. Aggen, C.G. Drake, B.J. Rini, Targeting PD-1 or PD-L1 in metastatic kidney cancer: combination therapy in the first-line setting, *Clin. Cancer Res.* 26 (9) (2020) 2087–2095.
- [23] K. Eddy, S. Chen, Overcoming immune evasion in melanoma, *Int. J. Mol. Sci.* 21 (23) (2020).
- [24] Y. Pico de Coana, M. Wolodarski, I. van der Haar Avila, T. Nakajima, S. Rentouli, A. Lundqvist, G. Masucci, J. Hansson, R. Kiessling, PD-1 checkpoint blockade in advanced melanoma patients: NK cells, monocytic subsets and host PD-L1 expression as predictive biomarker candidates, *Oncoimmunology* 9 (1) (2020), 1786888.
- [25] N. Shang, M. Figini, J. Shangguan, B. Wang, C. Sun, L. Pan, Q. Ma, Z. Zhang, Dendritic cells based immunotherapy, *Am. J. Cancer Res.* 7 (10) (2017) 2091–2102.
- [26] R. Ghinnagow, L.J. Cruz, E. Macho-Fernandez, C. Faveeuw, F. Trottein, Enhancement of adjuvant functions of natural killer T cells using nanovector delivery systems: application in anticancer immune therapy, *Front. Immunol.* 8 (2017) 879.
- [27] M.Y. Gerner, K.A. Casey, W. Kastenmuller, R.N. Germain, Dendritic cell and antigen dispersal landscapes regulate T cell immunity, *J. Exp. Med.* 214 (10) (2017) 3105–3122.
- [28] P.S. Hegde, D.S. Chen, Top 10 challenges in cancer immunotherapy, *Immunity* 52 (1) (2020) 17–35.
- [29] H. Raskov, A. Orhan, J.P. Christensen, I. Gogenur, Cytotoxic CD8⁺ T cells in cancer and cancer immunotherapy, *Br. J. Cancer* 124 (2) (2021) 359–367.
- [30] W. Zhang, L. Xu, H.B. Park, J. Hwang, M. Kwak, P.C.W. Lee, G. Liang, X. Zhang, J. Xu, J.O. Jin, *Escherichia coli* adhesion portion FimH functions as an adjuvant for cancer immunotherapy, *Nat. Commun.* 11 (1) (2020) 1187.
- [31] A. Marolda, K. Hunniger, S. Bottcher, W. Vivas, J. Löffler, M.T. Figge, O. Kurzai, Candida species-dependent release of IL-12 by dendritic cells induces different levels of NK cell stimulation, *J. Infect. Dis.* 221 (12) (2020) 2060–2071.
- [32] N. Oka, T. Markova, K. Tsuzuki, W. Li, Y. El-Darwish, M. Pencheva-Demireva, K. Yamanishi, H. Yamanishi, M. Sakagami, Y. Tanaka, H. Okamura, IL-12 regulates the expansion, phenotype, and function of murine NK cells activated by IL-15 and IL-18, *Cancer Immunol. Immunother.* 69 (9) (2020) 1699–1712.
- [33] S. Ghilas, M. Ambrosini, J.-C. Cance, M. Massé, H. Lelouard, M. Dalod, K. Crozat, NK Cells Orchestrate Splenic cDC1 Migration to Potentiate Antiviral Protective CD8⁺ T Cell Responses, *bioRxiv*, 2020.
- [34] G. Hodge, J. Barnawi, C. Jurisevic, D. Moffat, M. Holmes, P.N. Reynolds, H. Jersmann, S. Hodge, Lung cancer is associated with decreased expression of perforin, granzyme B and interferon (IFN)- γ by infiltrating lung tissue T cells, natural killer (NK) T-like and NK cells, *Clin. Exp. Immunol.* 178 (1) (2014) 79–85.
- [35] S. Bae, K. Oh, H. Kim, Y. Kim, H.R. Kim, Y.I. Hwang, D.S. Lee, J.S. Kang, W.J. Lee, The effect of alloferon on the enhancement of NK cell cytotoxicity against cancer via the up-regulation of perforin/granzyme B secretion, *Immunobiology* 218 (8) (2013) 1026–1033.

- [36] D.F. DeBlaker-Hohe, A. Yamauchi, C.R. Yu, J.A. Horvath-Arcidiacono, E.T. Bloom, IL-12 synergizes with IL-2 to induce lymphokine-activated cytotoxicity and perforin and granzyme gene expression in fresh human NK cells, *Cell. Immunol.* 165 (1) (1995) 33–43.
- [37] N. Zhu, X. Lv, Y. Wang, J. Li, Y. Liu, W. Lu, L. Yang, J. Zhao, F. Wang, L.W. Zhang, Comparison of immunoregulatory effects of polysaccharides from three natural herbs and cellular uptake in dendritic cells, *Int. J. Biol. Macromol.* 93 (Pt A) (2016) 940–951.
- [38] G. Pang, S. Zhang, X. Zhou, H. Yu, Y. Wu, T. Jiang, X. Zhang, F. Wang, Y. Wang, L.W. Zhang, Immunoactive polysaccharide functionalized gold nanocomposites promote dendritic cell stimulation and antitumor effects, *Nanomedicine (Lond)* 14 (10) (2019) 1291–1306.
- [39] C.J. Chen, Z.L. Li, Q. Fu, Y. Liu, X. Lei, H.C. Wu, Y.F. Liu, Effect of Astragalus polysaccharides on the phenotype and functions of human dendritic cells in vitro, *Nan Fang Yi Ke Da Xue Xue Bao* 29 (6) (2009) 1192–1194.
- [40] L. Zhou, Z. Liu, Z. Wang, S. Yu, T. Long, X. Zhou, Y. Bao, Astragalus polysaccharides exerts immunomodulatory effects via TLR4-mediated MyD88-dependent signaling pathway in vitro and in vivo, *Sci. Rep.* 7 (2017) 44822.
- [41] M. Mathieu, N. Labrecque, Murine superficial lymph node surgery, *J. Vis. Exp.* (63) (2012), e3444.
- [42] J.O. Jin, S. Jang, H. Kim, J. Oh, S. Shim, M. Kwak, P.C.W. Lee, Immunostimulatory agent evaluation: lymphoid tissue extraction and injection route-dependent dendritic cell activation, *J. Vis. Exp.* (139) (2018).
- [43] Q. Li, D. Wang, D. Bai, C. Cai, J. Li, C. Yan, S. Zhang, Z. Wu, J. Hao, G. Yu, Photoprotective effect of Astragalus membranaceus polysaccharide on UVA-induced damage in HaCaT cells, *PLoS One* 15 (7) (2020), e0235515.
- [44] W. Zhang, J. Hwang, H.B. Park, S.M. Lim, S. Go, J. Kim, I. Choi, S. You, J.O. Jin, Human peripheral blood dendritic cell and T cell activation by Codium fragile polysaccharide, *Mar. Drugs* 18 (11) (2020).
- [45] H.B. Park, J. Hwang, S.M. Lim, W. Zhang, J.O. Jin, Dendritic cell-mediated cancer immunotherapy with Ecklonia cava fucoidan, *Int. J. Biol. Macromol.* 159 (2020) 941–947.
- [46] W. Zhang, M. Kwak, H.B. Park, T. Okimura, T. Oda, P.C. Lee, J.O. Jin, Activation of human dendritic cells by Ascophyllum purified from Ascophyllum nodosum, *Mar. Drugs* 17 (1) (2019).
- [47] Y.C. Huang, H.J. Tsay, M.K. Lu, C.H. Lin, C.W. Yeh, H.K. Liu, Y.J. Shiao, Astragalus membranaceus-polysaccharides ameliorates obesity, hepatic steatosis, neuroinflammation and cognition impairment without affecting amyloid deposition in metabolically stressed APPswe/PS1dE9 mice, *Int. J. Mol. Sci.* 18 (12) (2017).
- [48] X. Jia, C. Zhang, J. Qiu, L. Wang, J. Bao, K. Wang, Y. Zhang, M. Chen, J. Wan, H. Su, J. Han, C. He, Purification, structural characterization and anticancer activity of the novel polysaccharides from Rhynchosia minima root, *Carbohydr. Polym.* 132 (2015) 67–71.
- [49] J. Hu, X. Jia, X. Fang, P. Li, C. He, M. Chen, Ultrasonic extraction, antioxidant and anticancer activities of novel polysaccharides from Chuanxiong rhizome, *Int. J. Biol. Macromol.* 85 (2016) 277–284.
- [50] L.F. Ballesteros, M.A. Cerqueira, J.A. Teixeira, S.I. Mussatto, Characterization of polysaccharides extracted from spent coffee grounds by alkali pretreatment, *Carbohydr. Polym.* 127 (2015) 347–354.
- [51] Y. Zheng, W. Ren, L. Zhang, Y. Zhang, D. Liu, Y. Liu, A review of the pharmacological action of Astragalus polysaccharide, *Front. Pharmacol.* 11 (2020) 349.
- [52] Y.C. Park, M.H. Kim, J.W. Kim, J.B. Kim, J.G. Lee, C.Y. Yu, S.H. Kim, I.M. Chung, J.K. Kim, R.N. Choi, J.D. Lim, Genotoxicity study of polysaccharide fraction from Astragalus membranaceus's aerial parts, *Toxicol. Res.* 30 (2) (2014) 131–138.
- [53] H. Abdel-Tawab, A. Abdel-Baki, A. El-Mallah, S. Al-Quraishy, H. Abdel-Haleem, In vivo and in vitro anticoccidial efficacy of Astragalus membranaceus against Eimeria papillata infection, *J. King Saud Univ. Sci.* 32 (3) (2020) 2269–2275.
- [54] J. Wang, J. Jia, L. Song, X. Gong, J. Xu, M. Yang, M. Li, Extraction, structure, and pharmacological activities of Astragalus polysaccharides, *Appl. Sci.* 9 (1) (2019) 122.
- [55] W. Zhang, T. Okimura, T. Oda, J.O. Jin, Ascophyllum induces activation of natural killer cells in mice in vivo and in vitro, *Mar. Drugs* 17 (4) (2019).
- [56] L. Xu, M. Kwak, W. Zhang, L. Zeng, P.C. Lee, J.O. Jin, Rehmannia glutinosa polysaccharide induces toll-like receptor 4 dependent spleen dendritic cell maturation and anti-cancer immunity, *Oncoimmunology* 6 (7) (2017), e1325981.
- [57] W. Zhang, T. Oda, Q. Yu, J.O. Jin, Fucoidan from Macrocystis pyrifera has powerful immune-modulatory effects compared to three other fucoidans, *Mar. Drugs* 13 (3) (2015) 1084–1104.
- [58] C. Oliveira, N.M. Neves, R.L. Reis, A. Martins, T.H. Silva, A review on fucoidan antitumor strategies: from a biological active agent to a structural component of fucoidan-based systems, *Carbohydr. Polym.* 239 (2020), 116131.
- [59] H.H. Lee, Y. Cho, G.H. Kim, H. Cho, Undaria pinnatifida fucoidan-rich extract recovers immunity of immunosuppressed mice, *J. Microbiol. Biotechnol.* 30 (3) (2020) 439–447.
- [60] W. Zhang, T. Okimura, L. Xu, L. Zhang, T. Oda, M. Kwak, Q. Yu, J.O. Jin, Ascophyllum functions as an adjuvant to promote anti-cancer effect by dendritic cell activation, *Oncotarget* 7 (15) (2016) 19284–19298.
- [61] L. Cohn, L. Delamarre, Dendritic cell-targeted vaccines, *Front. Immunol.* 5 (2014) 255.
- [62] C. Macri, E.S. Pang, T. Patton, M. O'Keeffe, Dendritic cell subsets, *Semin. Cell Dev. Biol.* 84 (2018) 11–21.
- [63] T. Worbs, S.I. Hammerschmidt, R. Förster, Dendritic cell migration in health and disease, *Nat. Rev. Immunol.* 17 (1) (2017) 30–48.
- [64] J.K. Ryu, S.J. Kim, S.H. Rah, J.I. Kang, H.E. Jung, D. Lee, H.K. Lee, J.O. Lee, B.S. Park, T.Y. Yoon, H.M. Kim, Reconstruction of LPS transfer cascade reveals structural determinants within LBP, CD14, and TLR4-MD2 for efficient LPS recognition and transfer, *Immunity* 46 (1) (2017) 38–50.
- [65] M. Maio, O. Hamid, J. Larkin, A. Cove, M. Altomonte, L. Calabro, S.A. Vardhana, C. Robert, R. Ibrahim, A. Anichini, J.D. Wolchok, A.M.D. Giacomo, Immune checkpoint inhibitors for cancer therapy in the COVID-19 era, *Clin. Cancer Res.* 26 (16) (2020) 4201–4205.
- [66] H. Herrscher, C. Robert, Immune checkpoint inhibitors in melanoma in the metastatic, neoadjuvant, and adjuvant setting, *Curr. Opin. Oncol.* 32 (2) (2020) 106–113.